

Efficient Mixture-of-Expert Language Model Inferencing Through Expert Sharing

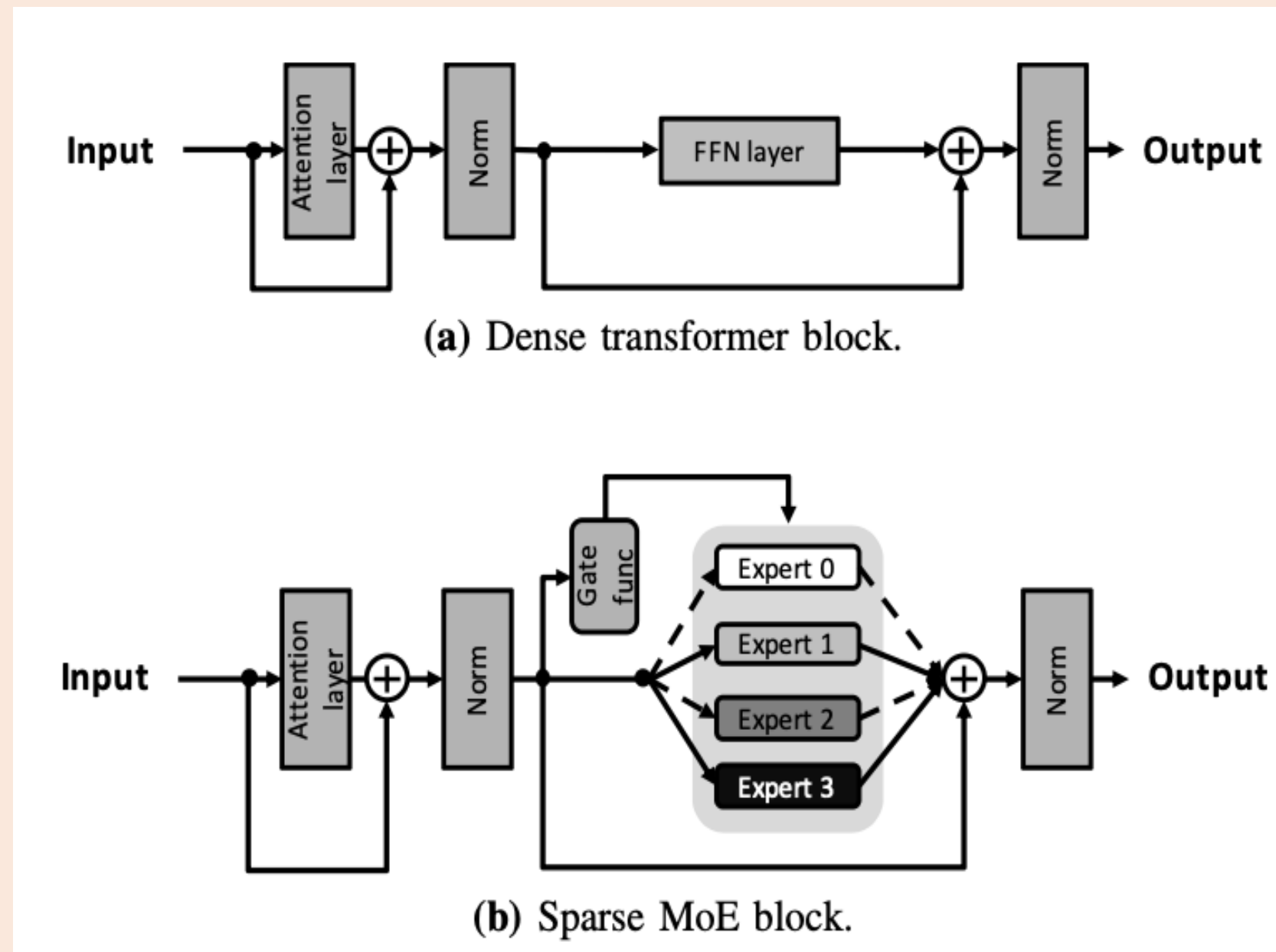
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1. Introduction

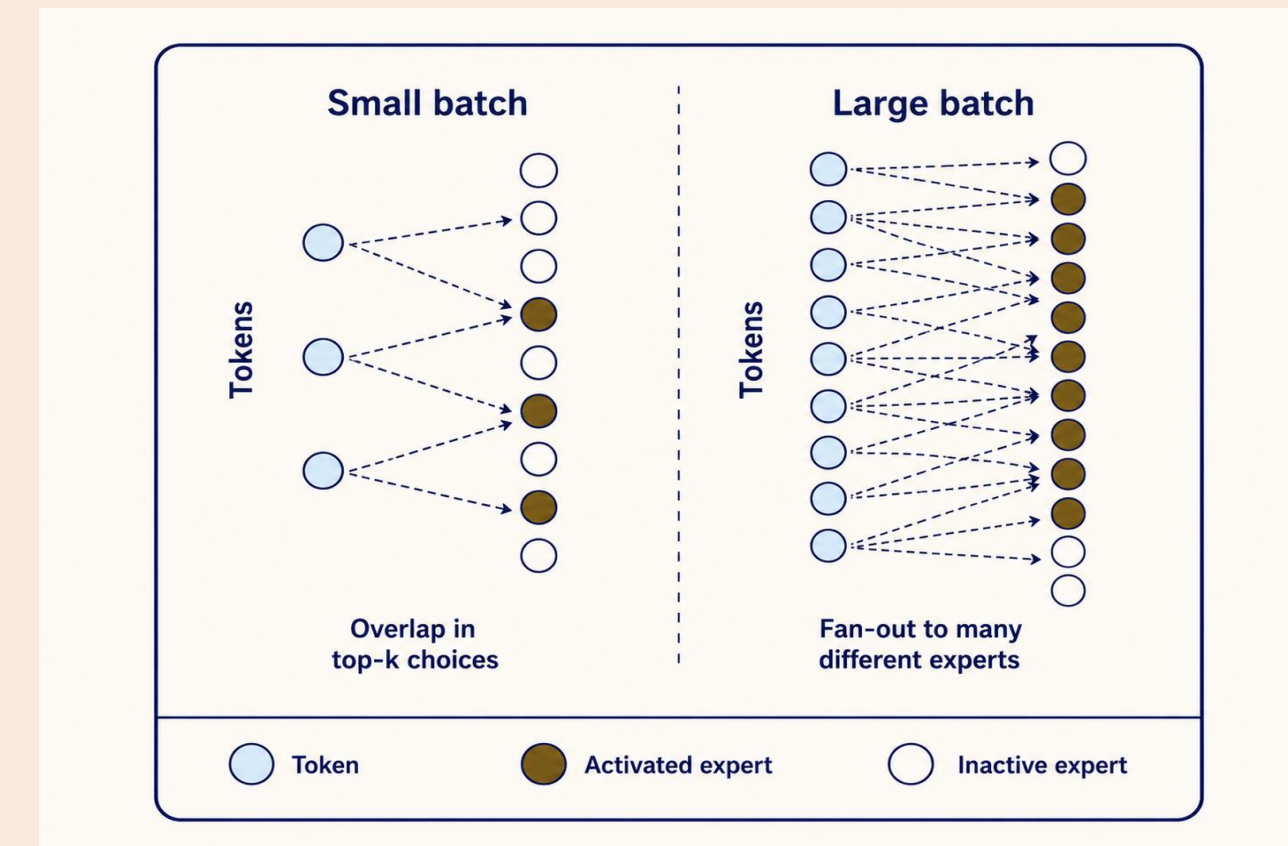
MoE LLMs use an innovative architecture that reduces the memory bandwidth required at inference time through a sparse block of **N experts**, replacing the large FFN block of dense models.

Tokens activate only a small subset of **k FFN experts** instead of the full dense FFN layer.



2. Background

The advantage of the sparse architecture is heavily reduced when running at moderate to high batch size as the superset of experts activated by a batch of tokens grows almost linearly with the size of the batch.



Some different strategies to share experts among different tokens have been developed. The state-of-the-art is **Opportunistic Expert Activation** (Oncescu et al., 2026). OEA preserves crucial top k experts per token, then piggybacks already-loaded batch experts to reduce unique activations and decode latency efficiently.

3. New methods

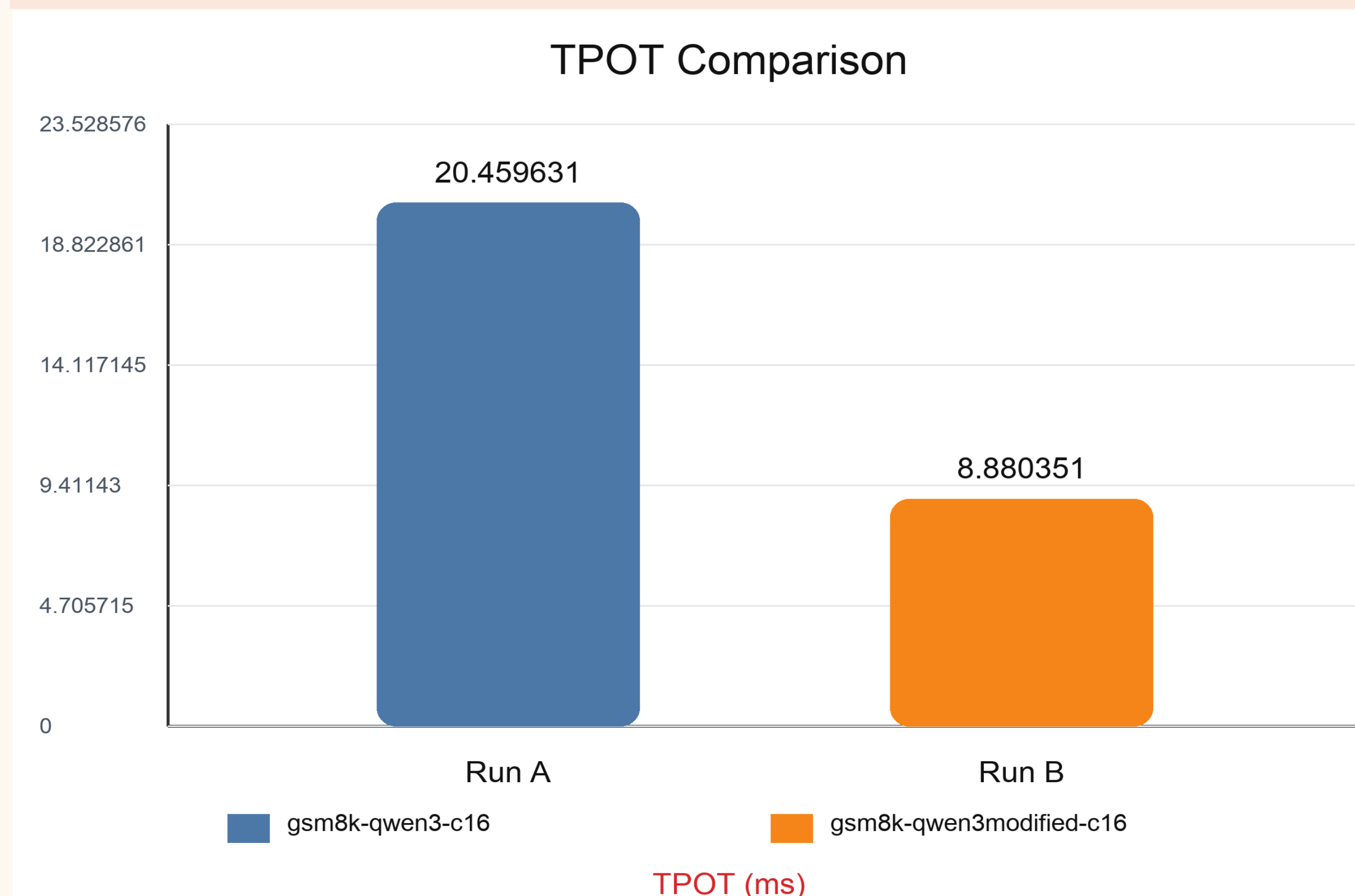
We profiled two families of models to verify which is the maximum achievable speedup on TPOT through extreme expert sharing.

Then we developed as a patch to SGLang two new heuristics to be tested:

- **No prize budget n(np)** -> starting from OEA prune further the superset of experts activated by a batch to reach a fixed expert budget
- **Fair regret relative lost core mass (rLCM) budget n** -> while pruning further from the superset of experts used by OEA we measure the penalization for each token as the fraction of top-k core mass lost after rerouting, then we reward them with greater routing influence in the next layer.

4 Profiling

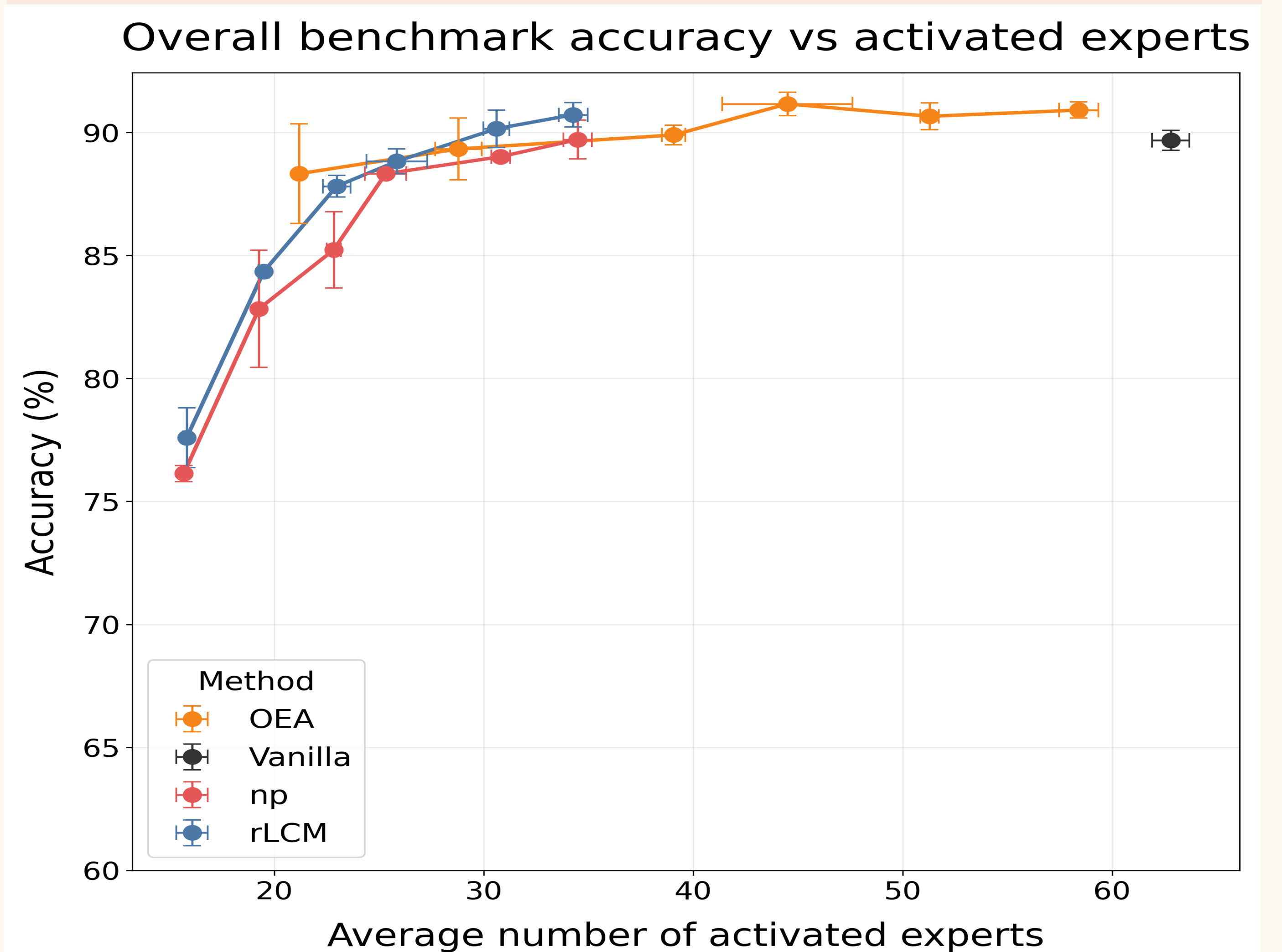
Expert sharing brought to its extreme on **Qwen3-30B-A3B** (all tokens share the same fixed experts) show a potential **TPOT speedup of 2.3x** under batch size 16 while running gsm8k benchmark. Similar results have been found also for Mixtral-8x7B(1.75x).



5. Accuracy vs number of experts activated

We tested OEA and our two new heuristics on a benchmark consisting of a set of 528 questions taken from different topics (96 of coding, 200 of general knowledge, 200 of math and 32 of biology and chemistry).

These questions were then sent to **Qwen3-30B-A3B** running on a NVIDIA-A100 with a **batch size 16**, while the loader shuffled them based on a random seed.



- All techniques reached the baseline accuracy (vanilla) with a clear reduction on the number of activated experts (2x reduction). This implies halving the time needed for the computation of the FFN block in an LLM model for each batch of tokens.
- All strategies are noisier than vanilla
- The two new heuristics challenge OEA on lower number of experts activated while maintaining the ability of the model to answer properly
- rLCM, which incorporates at layer L+1 what has happened at layer L is particularly consistent when the accuracy is already heavily degraded

Important Takeaways

1. Expert sharing is a powerful technique that can speedup up to 2.3x the TPOT
2. The speedup due to expert sharing is dependent on the model's architecture
3. Reducing too much the set of experts activated leads to an irrecoverable loss of accuracy
4. A key limit of expert sharing is the noise among runs